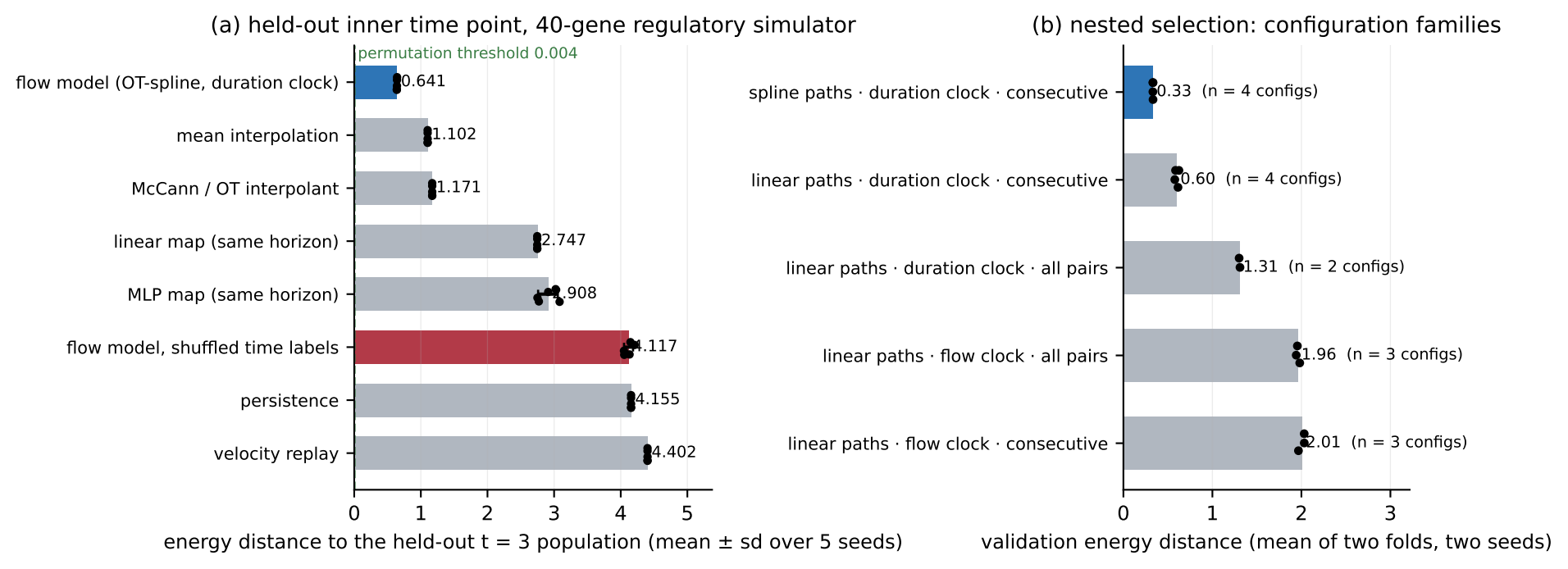
(a) held-out inner time point, 40-gene regulatory simulator
permutation threshold 0.004
flow model (OT-spline, duration clock)
mean interpolation
McCann / OT interpolant
linear map (same horizon)
MLP map (same horizon)
flow model, shuffled time labels
persistence
velocity replay
0.641
1.102
1.171
2.747
2.908
4.117
4.155
4.402
0
1
2
3
4
5
energy distance to the held-out t = 3 population (mean ± sd over 5 seeds)
(b) nested selection: configuration families
spline paths · duration clock · consecutive
linear paths · duration clock · consecutive
linear paths · duration clock · all pairs
linear paths · flow clock · all pairs
linear paths · flow clock · consecutive
0.33 (n = 4 configs)
0.60 (n = 4 configs)
1.31 (n = 2 configs)
1.96 (n = 3 configs)
2.01 (n = 3 configs)
0
1
2
3
validation energy distance (mean of two folds, two seeds)